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PythonicDISORT: A Python reimplementation of the Discrete Ordinate Radiative Transfer package DISORT

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Software

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Summary

The Radiative Transfer Equation (RTE) models the processes of absorption, scattering and emission as electromagnetic radiation propagates through a medium. Consider a plane-parallel, horizontally homogeneous atmosphere with vertical coordinate τ (optical depth) increasing from top to bottom and directional coordinates ϕ for the azimuthal angle (positive is counterclockwise) and $\mu = \cos\theta$ for the polar direction (θ is the polar angle measured from the surface normal), with $\mu > 0$ pointing up following the convention of (K. Stamnes et al., 1988). Given three possible sources: blackbody emission from the atmosphere $s(\tau)$, scattering from a collimated beam of starlight with intensity I_0 and incident cosine polar and azimuthal angles μ_0, ϕ_0, and/or radiation from other atmospheric layers or the Earth's surface which is modeled by Dirichlet boundary conditions, the diffuse intensity $u(\tau, \mu, \phi)$ propagating in direction (μ, ϕ) is described by the 1D RTE (Chandrasekhar, 1960; K. Stamnes et al., 1988):

$$\begin{aligned}\mu \frac{\partial u(\tau,\mu,\phi)}{\partial \tau} = u(\tau,\mu,\phi) - \frac{\omega}{4\pi}\int_{-1}^{1}\int_{0}^{2\pi} p\left(\mu,\phi;\mu',\phi'\right) u\left(\tau,\mu',\phi'\right) \mathrm{d}\phi' \mathrm{d}\mu' \\ - \frac{\omega I_0}{4\pi} p\left(\mu,\phi;-\mu_0,\phi_0\right) \exp\left(-\mu_0^{-1}\tau\right) - s(\tau)\end{aligned} \tag{1}$$

Here ω is the single-scattering albedo and p the scattering phase function. These are assumed to be independent of τ, i.e. homogeneous in the atmospheric layer. An atmosphere with τ-dependent ω and p can be modelled by a multi-layer atmosphere with different ω and p for each layer.

The RTE is important in many fields of science and engineering, for example, in the retrieval of optical properties from measurements (McGuire et al., 2008; Teng et al., 2020; Torricella et al., 1999). The gold standard for numerically solving the 1D RTE is the Discrete Ordinate Radiative Transfer package `DISORT` which was coded in FORTRAN 77 and first released in 1988 (K. Stamnes et al., 1988; S. Stamnes, 1999). It has been widely used, for example by `MODTRAN` (Berk et al., 2014), `Streamer` (Key & Schweiger, 1998), and `SBDART` (Ricchiazzi et al., 1998), all of which are comprehensive radiative transfer models that are themselves widely used in atmospheric science, and by the three retrieval papers Torricella et al. (1999); McGuire et al. (2008); Teng et al. (2020). `DISORT` implements the Discrete Ordinates Method which has two key steps. First, the diffuse intensity function u and phase function p are expanded as the Fourier cosine series and Legendre series respectively:

$$\begin{aligned} u\left(\tau,\mu,\phi\right) &\approx \sum_{m=0} u^m\left(\tau,\mu\right)\cos\left(m\left(\phi_0-\phi\right)\right) \\ p\left(\mu,\phi;\mu',\phi'\right) = p\left(\cos\gamma\right) &\approx \sum_{\ell=0}(2\ell+1) g_\ell P_\ell\left(\cos\gamma\right)\end{aligned}$$

where γ is the scattering angle. These address the ϕ' integral in (1) and decompose the problem into solving

$$\mu \frac{du^m(\tau,\mu)}{d\tau} = u^m(\tau,\mu) - \int_{-1}^{1} D^m(\mu,\mu')\, u^m(\tau,\mu')\, \mathrm{d}\mu' - Q^m(\tau,\mu) - \delta_{0m} s(\tau)$$

for each Fourier mode of u. The terms D^m are derived from p and are thus also independent of τ. The second key step is to discretize the μ' integral using some quadrature scheme; `DISORT` uses the double-Gauss quadrature scheme from Sykes (1951). This results in a system of ordinary differential equations that can be solved using standard methods.

Our package `PythonicDISORT` is a Python 3 reimplementation of `DISORT` that replicates most of its functionality while being easier to install, use, and modify, though at the cost of computational speed. It has `DISORT`'s main features: multi-layer solver, delta-M scaling, Nakajima-Tanaka (NT) corrections, only flux option, direct beam source, isotropic internal source (blackbody emission), Dirichlet boundary conditions (diffuse flux boundary sources), Bi-Directional Reflectance Function (BDRF) for surface reflection, and more. In addition, `PythonicDISORT` has been tested against `DISORT` on `DISORT`'s own test problems. While packages that wrap `DISORT` in Python already exist (Connour & Wolff, 2020; Hu, 2017), `PythonicDISORT` is the first time `DISORT` has been reimplemented from scratch in Python.

Statement of need

`PythonicDISORT` is not meant to replace `DISORT`. Due to fundamental differences between Python and FORTRAN, `PythonicDISORT`, though optimized, remains about an order of magnitude slower than `DISORT`. Thus, projects that prioritize computational speed should still use `DISORT`. I will continue to optimize `PythonicDISORT`; there remain avenues for code vectorization among other optimizations. It is unlikely that `PythonicDISORT` can be optimized to achieve the speed of `DISORT` though. In addition, `PythonicDISORT` currently lacks `DISORT`'s latest features, most notably its pseudo-spherical correction, though I am open to adding new features and I added a subroutine to compute actinic fluxes to satisfy a user request.

`PythonicDISORT` is instead designed with three goals in mind. First, it is meant to be a pedagogical and exploratory tool. `PythonicDISORT`'s ease of installation and use makes it a low-barrier introduction to Radiative Transfer and Discrete Ordinates Solvers. Even researchers who are experienced in the field may find it useful to experiment with `PythonicDISORT` before deciding whether and how to upscale with `DISORT`. Installation of `PythonicDISORT` through `pip` should be system agnostic as `PythonicDISORT`'s core dependencies are only `NumPy` (Harris et al., 2020) and `SciPy` (Virtanen et al., 2020). I also intend to implement `conda` installation. In addition, using `PythonicDISORT` is as simple as calling the Python function `pydisort`. In contrast, `DISORT` requires FORTRAN compilers, has a lengthy and system dependent installation, and each call requires shell script for compilation and execution.

Second, `PythonicDISORT` is designed to be modified by users to suit their needs. Given that Python is a widely-used high-level language, `PythonicDISORT`'s code should be understandable, at least more so than `DISORT`'s FORTRAN code. Moreover, `PythonicDISORT` comes with a Jupyter Notebook (Kluyver et al., 2016) – our *Comprehensive Documentation* – that breaks down both the mathematics and code behind the solver. Users can in theory follow the Notebook to recode `PythonicDISORT` from scratch; it should at least help them make modifications.

Third, `PythonicDISORT` is intended to be a testbed. For the same reasons given above, it should be easier to implement and test experimental features in `PythonicDISORT` than in `DISORT`. This should expedite research and development for `DISORT` and similar algorithms.

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PythonicDISORT was first released on PyPI and GitHub on May 30, 2023. It was used in Ho & Pincus (2024) and is being used in at least three ongoing projects: on the Two-Stream Approximations, on atmospheric photolysis, and on the topographic mapping of Mars through photoclinometry. I will continue to maintain and upgrade PythonicDISORT. The latest version: PythonicDISORT v0.9.3 was released on October 13, 2024.

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